

Project III: Employee Retention Analytics

Title: Understanding Workforce Attrition Through Statistical Analysis

Phase 1 — Problem formulation and data understanding

Task 1: Problem Definition

Real-World Problem

Employee turnover is a critical issue for organizations because it results in increased operational costs, reduced productivity, and loss of experienced talent. When employees leave, organizations must invest time and money in recruiting, hiring, and training replacements. High turnover can also negatively impact remaining employees through increased workloads and decreased morale.

The real-world problem addressed in this project is identifying the key factors that cause employees to leave their jobs. Without understanding why employees resign, organizations struggle to implement effective retention strategies and risk continued workforce instability.

Importance of the Problem

This problem is important because retaining employees is significantly more cost-effective than replacing them. High turnover affects organizational performance, employee engagement, and long-term growth. By using data-driven analysis, organizations can move beyond assumptions and identify measurable factors—such as salary, job satisfaction, workload, and tenure—that influence employee turnover.

Understanding these factors allows organizations to design targeted interventions that improve employee satisfaction, reduce burnout, and increase retention.

Stakeholders

The primary stakeholders affected by employee turnover include:

- Human Resources departments, responsible for recruitment, retention, and workforce planning
- Managers and supervisors, who must manage team performance and staffing continuity
- Employees, whose morale and workload are impacted by turnover
- Executive leadership, responsible for controlling costs and maintaining organizational performance

Decisions Informed by the Results

The findings from this research can support the following organizational decisions:

- Adjusting salary and compensation structures
- Improving employee satisfaction and engagement initiatives
- Managing workloads and project assignments more effectively
- Identifying high-risk employees early and implementing retention strategies

Purpose of the Study

The purpose of this study is to analyze employee retention data using statistical methods to determine which factors most strongly influence employee turnover. By examining variables such as salary level, job satisfaction, workload, tenure, promotions, and workplace conditions, this research aims to provide actionable insights that organizations can use to reduce attrition and improve employee retention.

Task 2: Dataset Investigation

Dataset Profile Report

The dataset used for this project is the HR Employee Retention Dataset, obtained from Kaggle. This dataset is publicly available and commonly used for analyzing employee turnover and retention patterns. Kaggle serves as the platform hosting the dataset, while the data represents simulated human resources records designed for analysis and learning purposes.

Dataset Overview

The dataset contains information on approximately a few thousand employees and includes multiple variables related to employee performance, workload, compensation, and employment history. Each row represents an individual employee, while each column represents a specific employee attribute.

The dataset is structured, tabular, and well-suited for statistical analysis and data-driven decision-making.

Variables Description

Key variables in the dataset include:

- Satisfaction level – measures employee job satisfaction
- Last evaluation – employee performance score
- Number of projects – total projects assigned
- Average monthly hours – number of hours worked per month
- Time spent at company – number of years employed
- Work accident – indicates if an employee experienced a work accident
- Promotion in last 5 years – reflects promotion history
- Salary – salary category (low, medium, high)
- Left – indicates whether an employee left the company

These variables represent both numerical data (e.g., working hours, evaluation scores) and categorical data (e.g., salary levels and department), making the dataset suitable for multiple types of statistical analysis.

Data Collection Process

The dataset represents employee information typically collected by Human Resources departments. Data such as satisfaction levels, workload, performance evaluations, promotions, and salary categories are usually gathered from internal HR systems, performance tracking tools, and employee surveys.

Although the dataset is simulated, it reflects realistic workplace data and accurately represents typical HR data collection practices used in organizations.

Relevance to the Study

This dataset is highly relevant for analyzing employee turnover because it includes key factors that influence retention, such as job satisfaction, salary, workload, tenure, and promotion history. These variables allow for meaningful comparisons between employees who stayed and those who left the organization.

As a result, the dataset supports both descriptive and inferential statistical analysis and is useful for identifying trends and making data-driven recommendations for improving employee retention.

Task 3: Research questions

Salary & Compensation

1. How does salary level influence employee turnover rates across different compensation groups?
2. To what extent do lower salary levels contribute to increased employee attrition compared to higher salary groups?

Job Satisfaction

3. How does job satisfaction level influence the likelihood of employee turnover?
4. At what satisfaction threshold does employee turnover begin to significantly increase?
5. How do employees with high performance but low satisfaction differ in turnover rates compared to other groups?

Workload & Productivity

6. How do monthly working hours influence employee turnover rates, particularly at extreme workload levels?
7. What impact does the number of projects assigned have on employee turnover across different workload groups?
8. Is there a critical workload threshold where employee turnover rates increase significantly?

Tenure & Career Growth

9. How does time spent at the company influence employee turnover rates at different stages of employment?
10. At what point is an employee's tenure turnover most likely to occur?

Organizational Factors

11. How have promotions within the last five years influenced employee retention and turnover rates?
12. What is the relationship between workplace factors (such as work accidents) and employee turnover?
13. Which combinations of employee characteristics (salary, satisfaction, workload, and tenure) are most strongly associated with high turnover risk?